

Chapter 4: Acid-Base Equilibrium and Buffer Systems

The mechanism by which buffer solutions resist changes in pH depends on the equilibrium between a weak acid and its conjugate base. When a strong acid is added to the buffer, the conjugate base reacts with the excess protons, shifting the equilibrium and minimizing the resulting change in pH.

A common trap students fall into is assuming buffer capacity is unlimited. In reality, once the weak acid or conjugate base is fully consumed, the buffer system loses its ability to resist pH changes, and the solution pH will change rapidly with further addition of acid or base.

Clinically, this principle explains why the bicarbonate buffer system in blood plasma is critical: small disruptions in the ratio of bicarbonate to carbonic acid can cause significant changes in blood pH, leading to acidosis or alkalosis. Diagnosis of these conditions often requires interpreting arterial blood gas values relative to the Henderson-Hasselbalch equation.

Why this matters: the DAT and MCAT both test buffer calculations heavily, and understanding the underlying mechanism (not just memorizing the formula) is the key to answering multi-step equilibrium questions correctly.